

X: Generative Models in 3D

3D CV

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In the Previous Episode

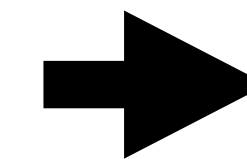
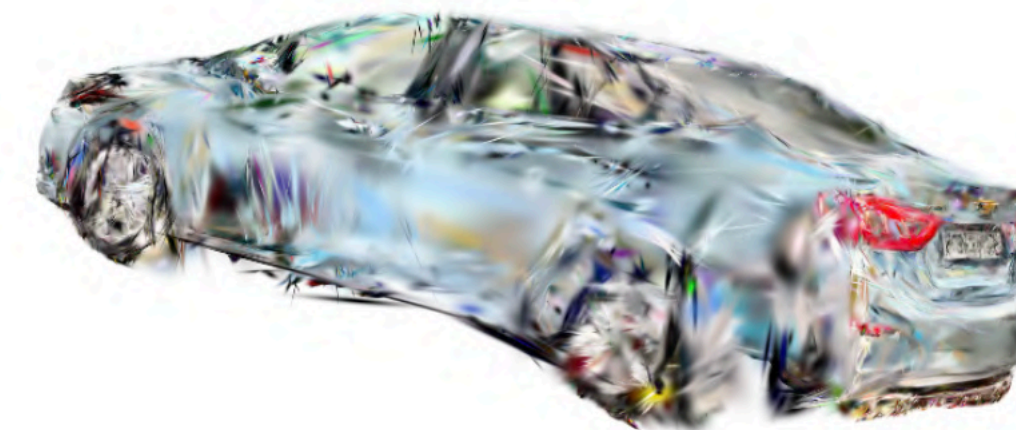
- Novel view synthesis
 - Input: posed images of a scene
 - Goal: render the scene from novel viewpoints
- Approaches based on volumetric representations
 - Neural Radiance Fields
 - Gaussian Splatting
- Works great for carefully-collected data



What if There's no Data?

Or Data is not Good Enough

- Novel view synthesis is prone to failures due to data limitations
- Diffusion models excel at photorealistic image generation
- Can we leverage these models to generate 3D data?
 - From scratch
 - To improve reconstruction



Diffusion in a Nutshell

Diffusion: The Idea

- **Forward process:** gradually add Gaussian noise to data, controlled by timestep t :

$$x_t = \alpha_t x_0 + \sigma_t \mathcal{E}$$

- **Reverse process:** learn a network $x_\phi(x_t, t)$ that predicts the clean sample x_0 from x_t
- **Generation:** start from pure noise, iteratively denoise step by step, each step refines the estimate, gradually revealing structure

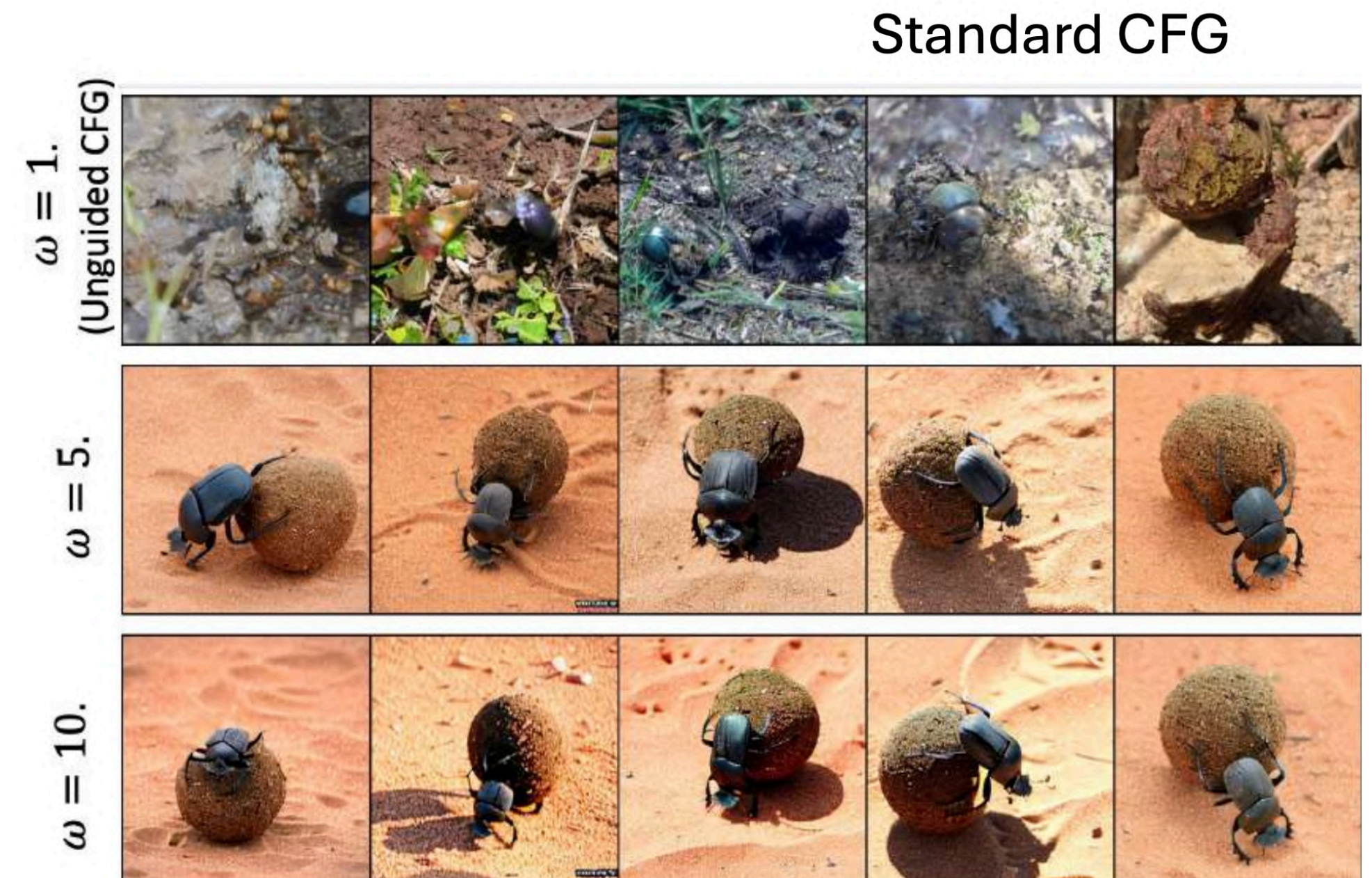


Classifier-Free Guidance

- Diffusion models allow conditional generation: add additional information to help model restore the original image
- At inference we blend conditional and unconditional predictions

$$x_{\phi} = x_{\phi}^{\emptyset} + s \cdot (x_{\phi}^c - x_{\phi}^{\emptyset})$$

- Here x_{ϕ}^c and $x_{\phi}^{\emptyset}$ are the prediction with and without condition respectively, whereas scale s controls the condition influence



From Text-to-2d to Text-to-3D

DreamFusion

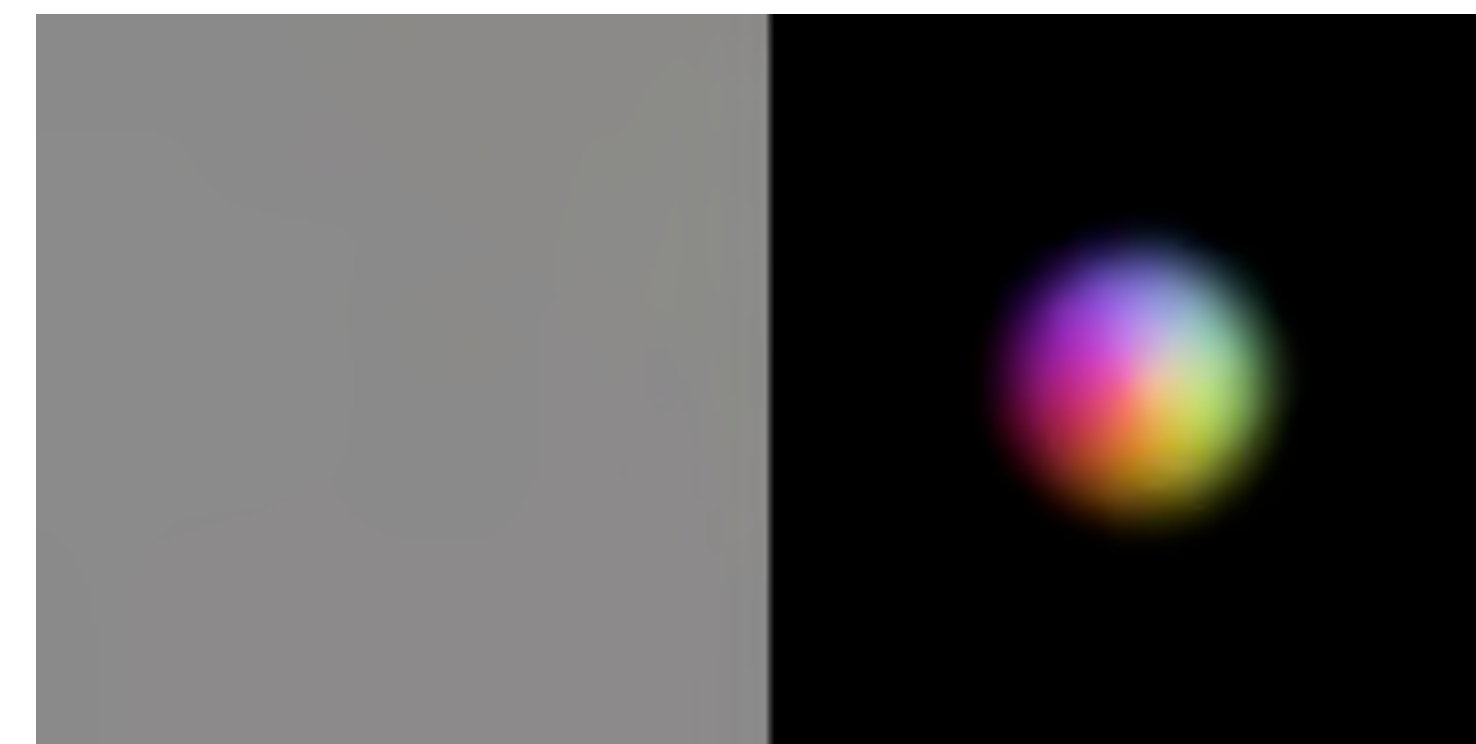
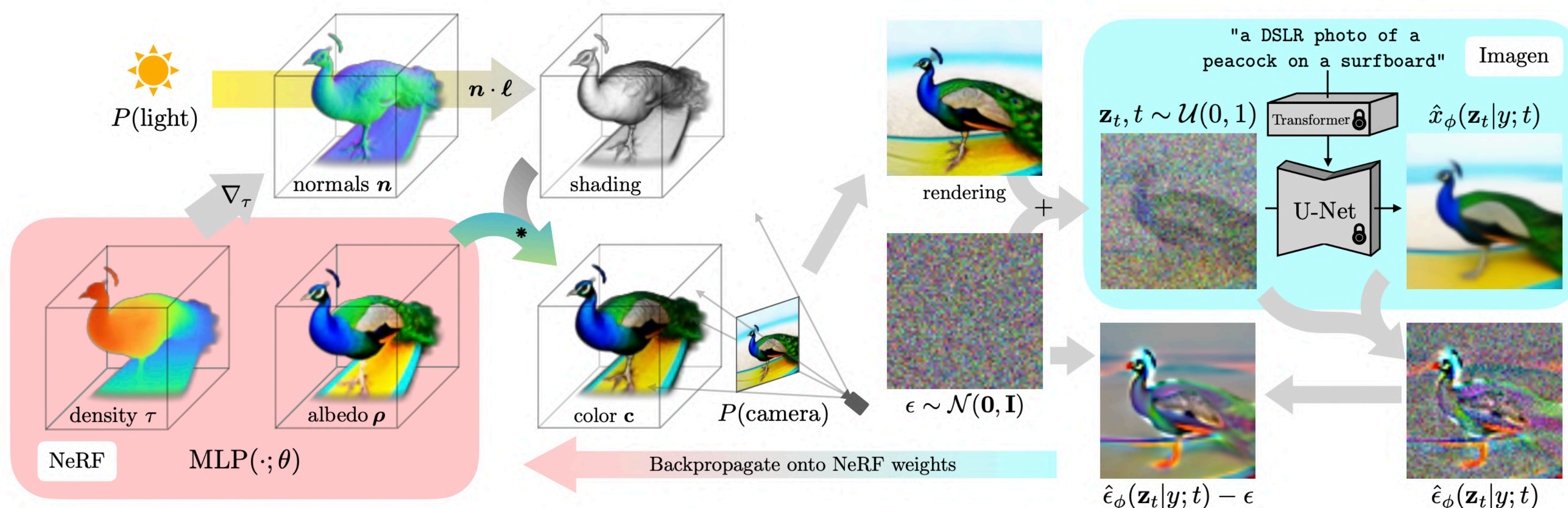
- Pre-trained text-to-2D diffusion model
- Input: text prompt
 - “A DSLR photo of a squirrel wearing a kimono reading a book”
- Output: 3D model
- Parameterise 3D model with NeRF $x(\theta)$
- Without ground truth images, what would be the objective?



Score Distillation Sampling (SDS)

- Render a NeRF based image $x_0(\theta)$ and corrupt it with noise $x_t = \alpha_t x_0 + \sigma_t \epsilon$
- Estimate the uncorrupted image with the diffusion model $x_\phi(x_t, t)$
- Define a “pseudo-objective”: $\nabla_\theta \mathcal{L}_{SDS}(\theta) = \mathbb{E}_{t,\epsilon} w(t) \left(x_\phi(x_t, t) - x_0(\theta) \right) \frac{\partial x_0}{\partial \theta} \approx \nabla_\theta \mathbb{E}_{t,\epsilon} \|x_\phi - x_0(\theta)\|^2$

(depends on θ)



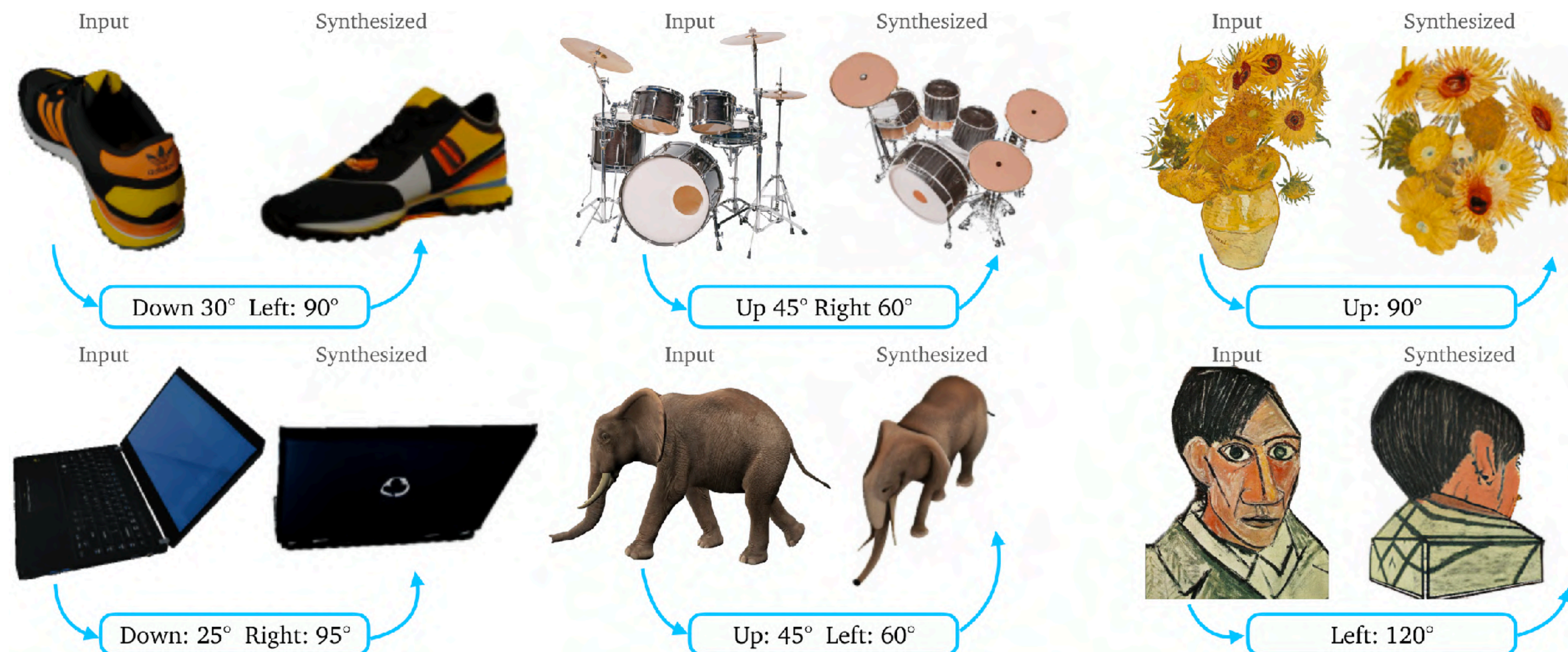
DreamFusion: Results

- Pros:
 - It works!
- Cons:
 - ... and sometimes it does not
 - Low sample diversity due to mode-seeking objective
 - Janus problem: 2D model tries to treat all sides of the model equally



Camera-conditioned Models

- Diffusion models excel at conditional generation
- Zero-1-to-3: we can fine-tune a 2D model to predict novel views in 3D on synthetic data



Distilling Camera-Conditioned Models in 3D

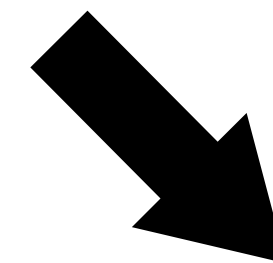
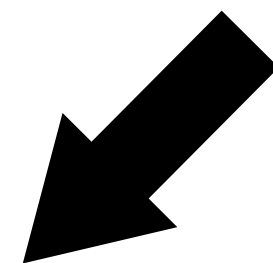
- Novel-views can be inconsistent across samples
- We can use the camera-conditioned model as a backbone for SDS and distill generated images in a consistent 3D model



The Fork

- DreamFusion showed that 2D diffusion knows a lot about 3D, but SDS alone is fragile, slow, and limited
- As a result, the field split into two research directions

DreamFusion



Train dedicated 3D generative models to generate assets directly

Move away from 2D diffusion, train on large-scale synthetic 3D data

Use diffusion as an engine for novel view synthesis to reconstruct real scenes

Keep 2D diffusion, fine-tune on real multi-view data, generate new views instead of SDS

3D Generation on Synthetic Data

From Image-Based Models to Native 3D Diffusions

Objaverse & Objaverse-XL

- Contain 800k & 10m+ 3D assets correspondingly
- Quality varies widely; most pipelines filter down to 300–500K usable assets
- Sufficient to train dedicated 3D generation models for the first time
- Synthetic data leads to synthetic appearance

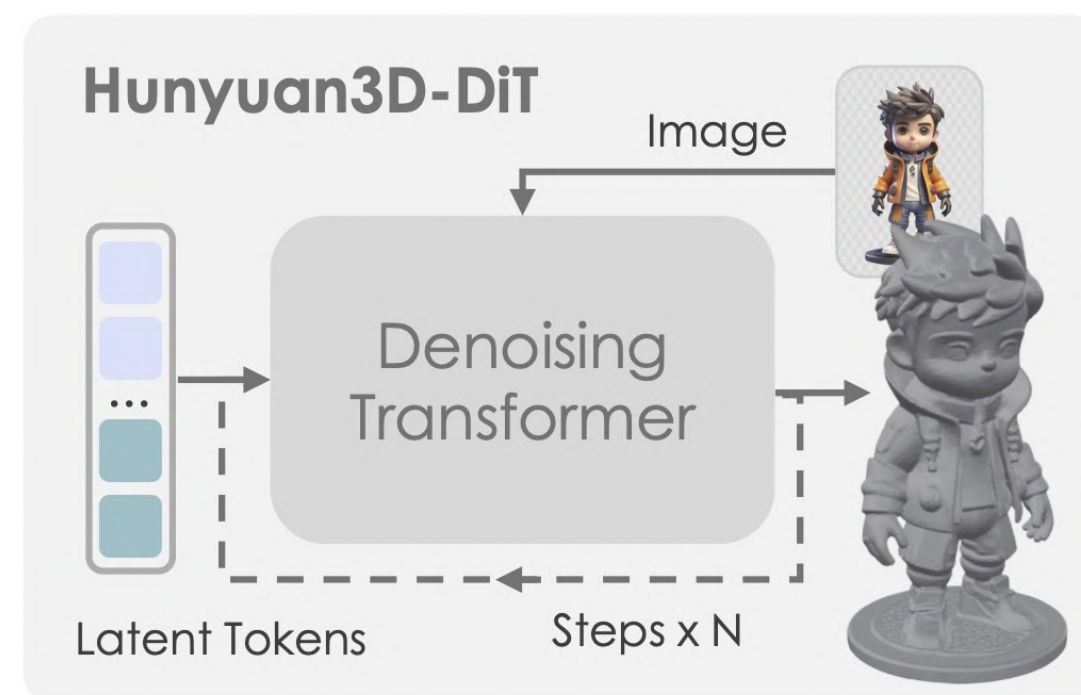
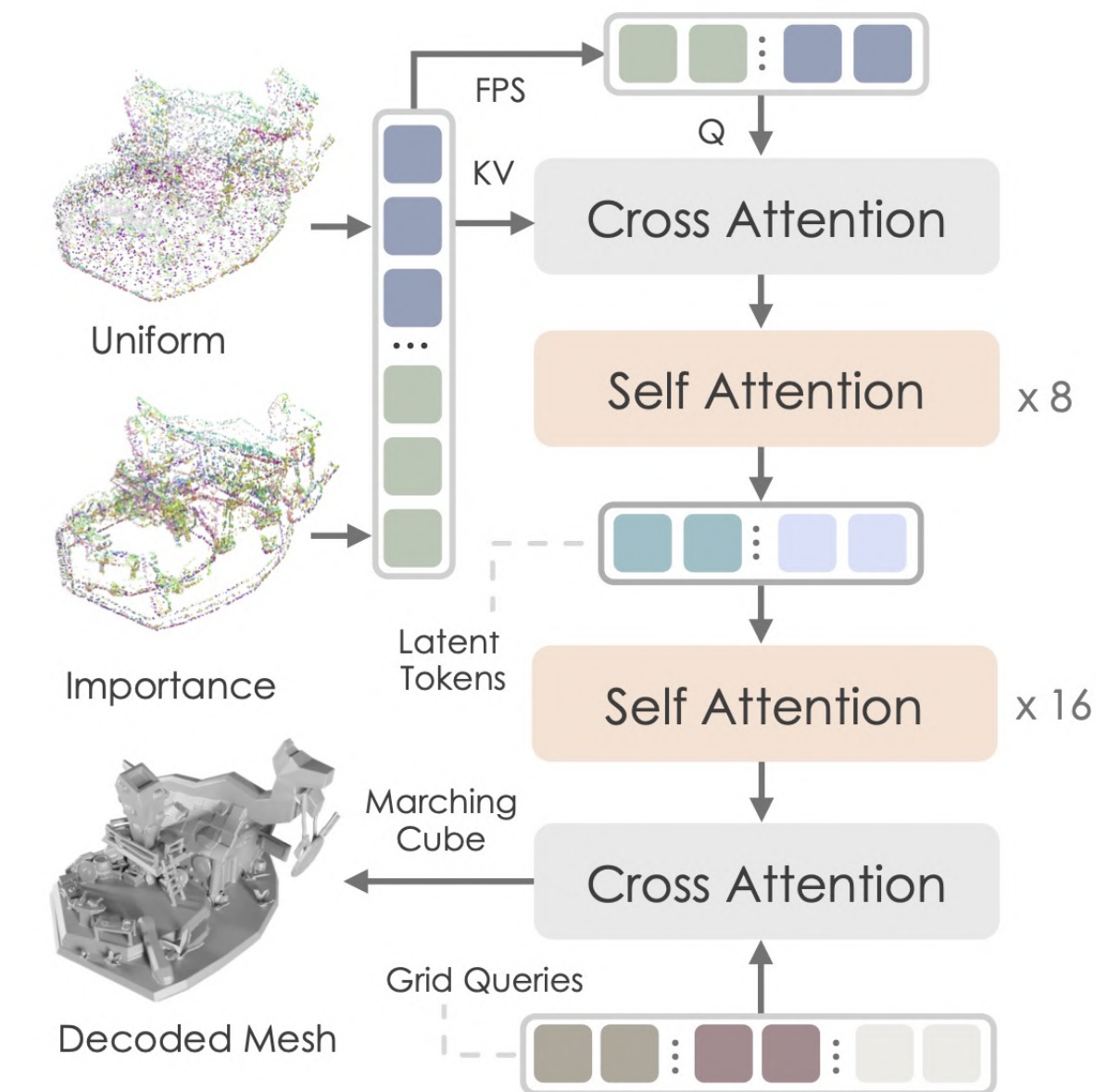


Hunyan3D (1/2)

- Split task into shape & texture generation
- Dedicated latent diffusion shape model
- Texture generation based on multi-view diffusion

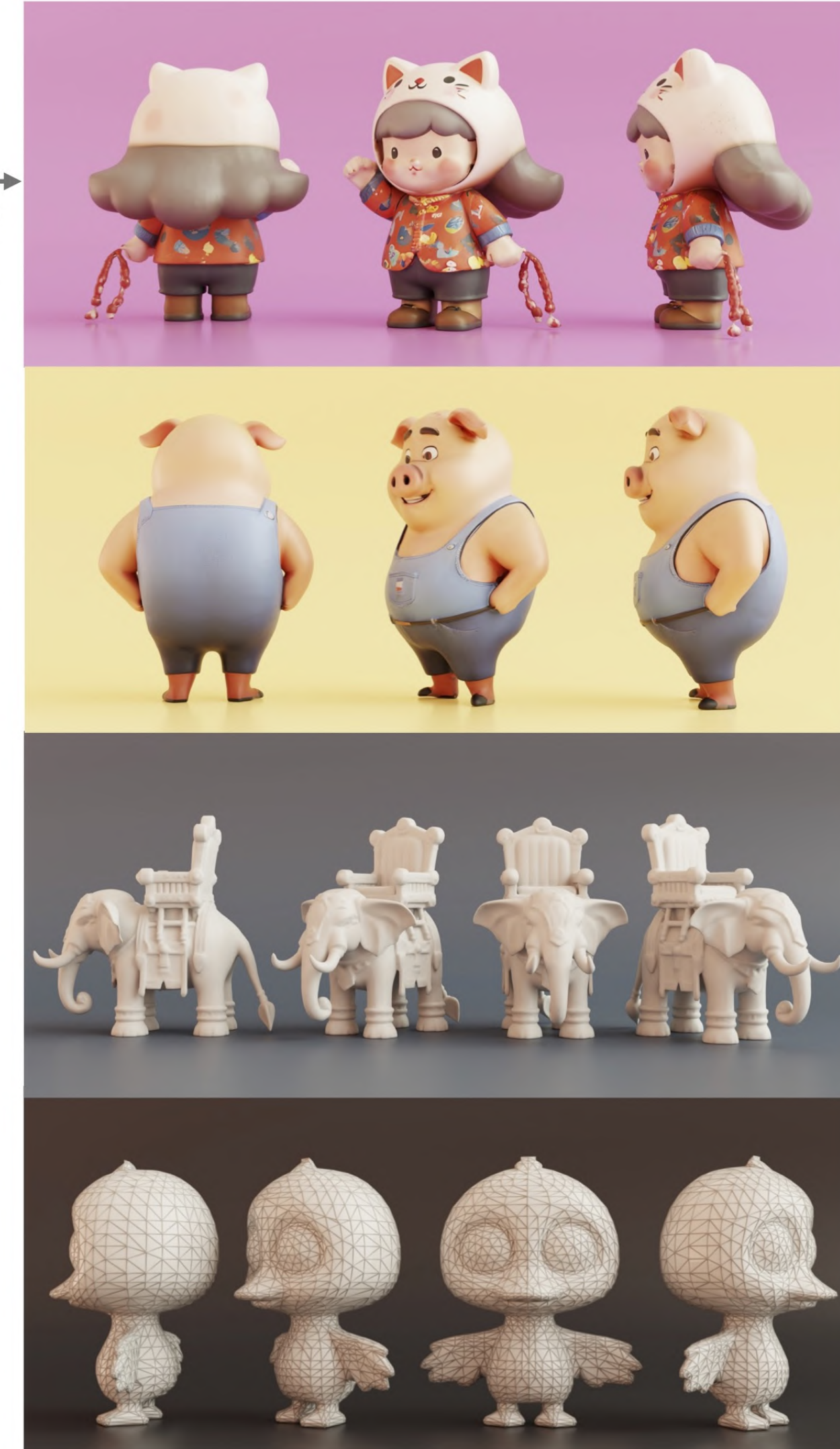
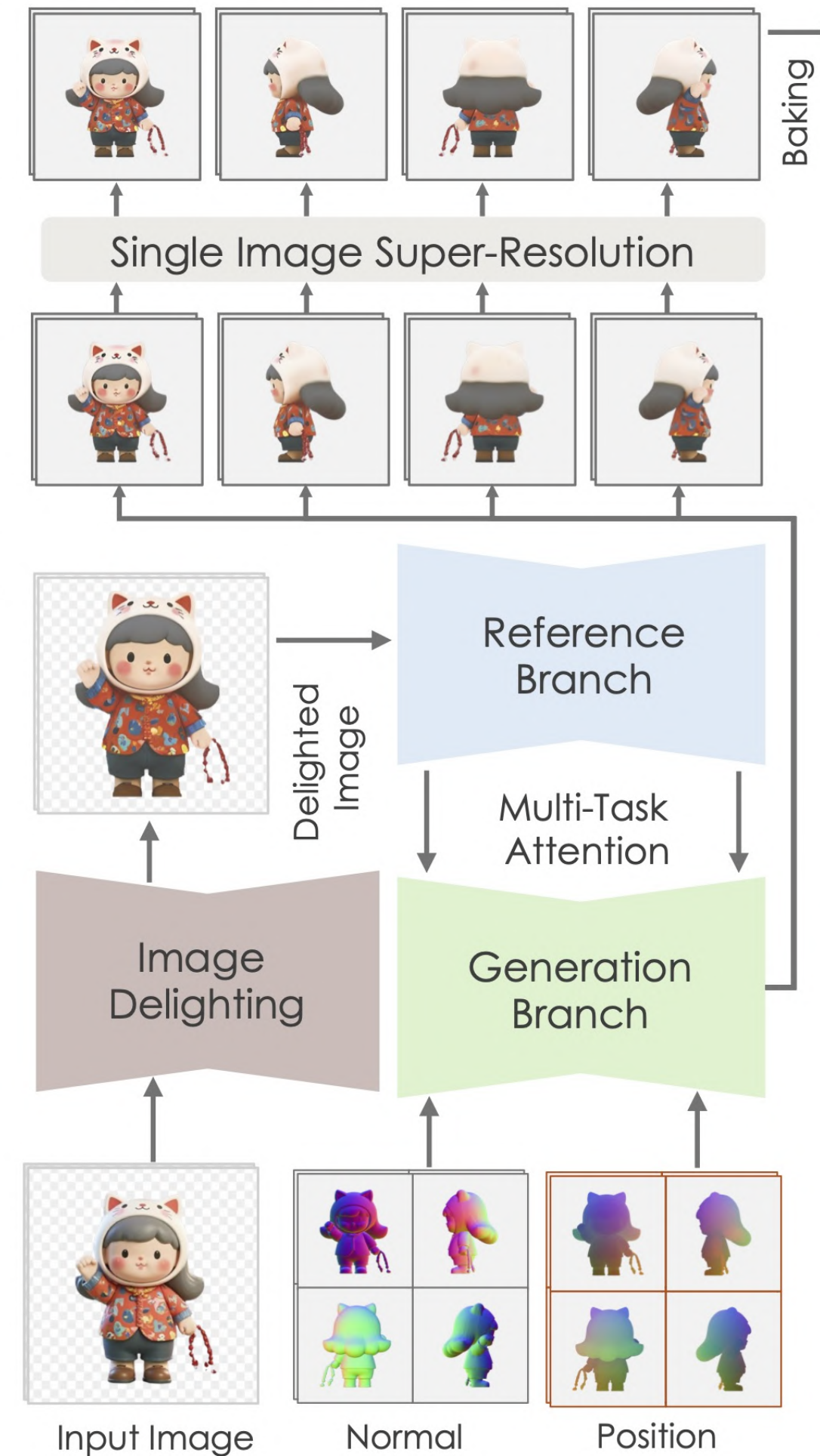
Shape Generation

Hunyuan3D-ShapeVAE



Texture Synthesis

Hunyuan3D-Paint



Trellis (1/2)

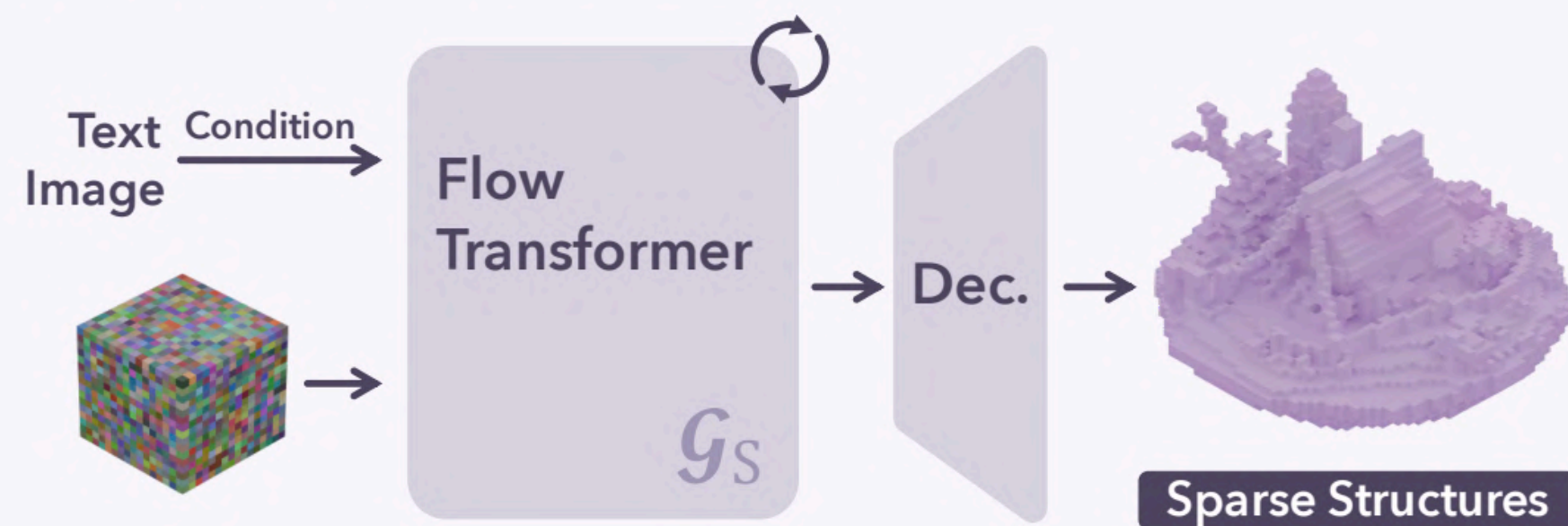
3D Assets Encoding & Decoding

Structured Latent Representation Learning

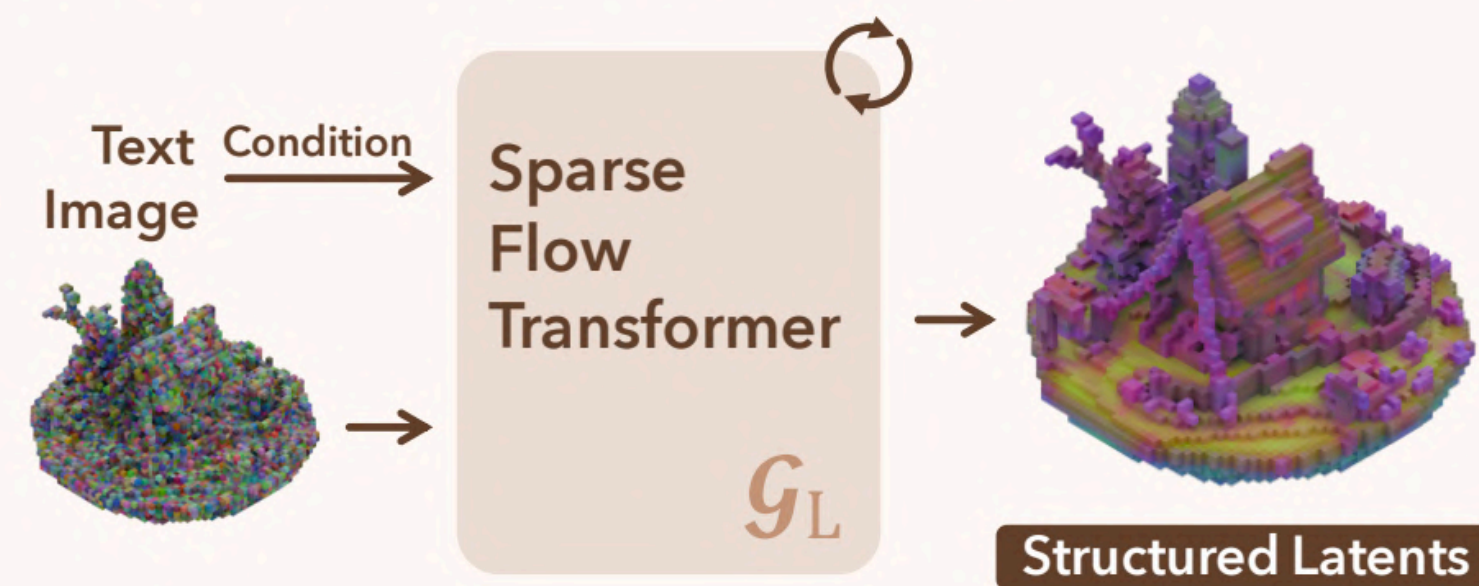


3D Assets Generation

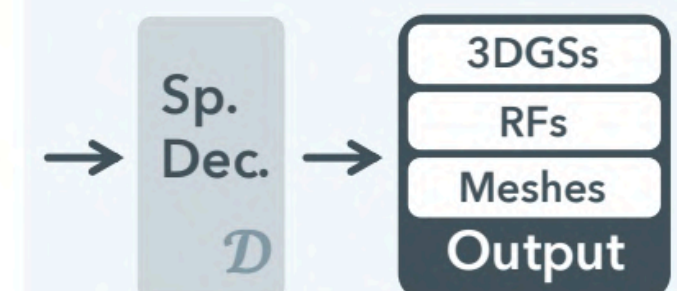
Structure Generation



Structured Latents Generation

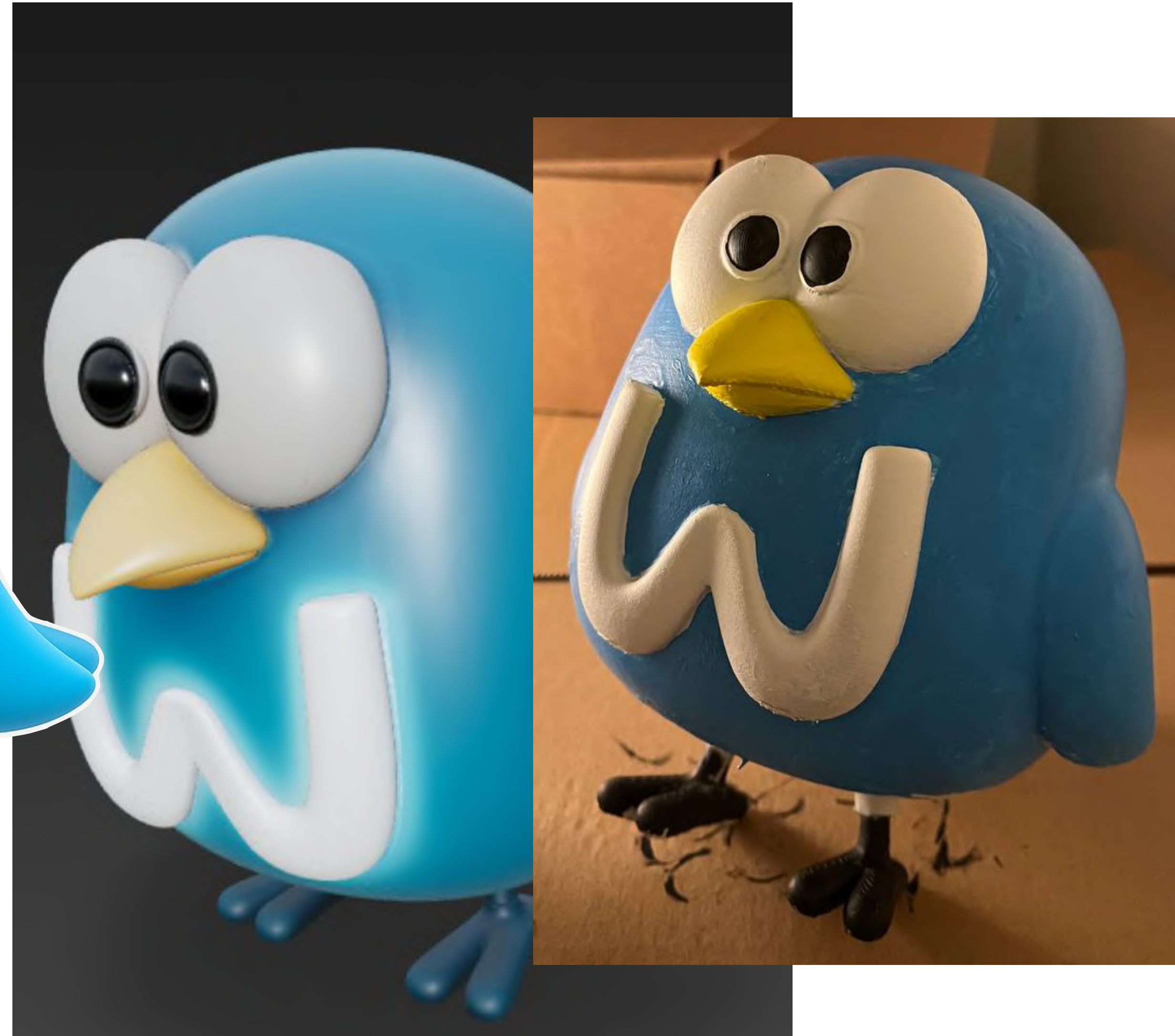
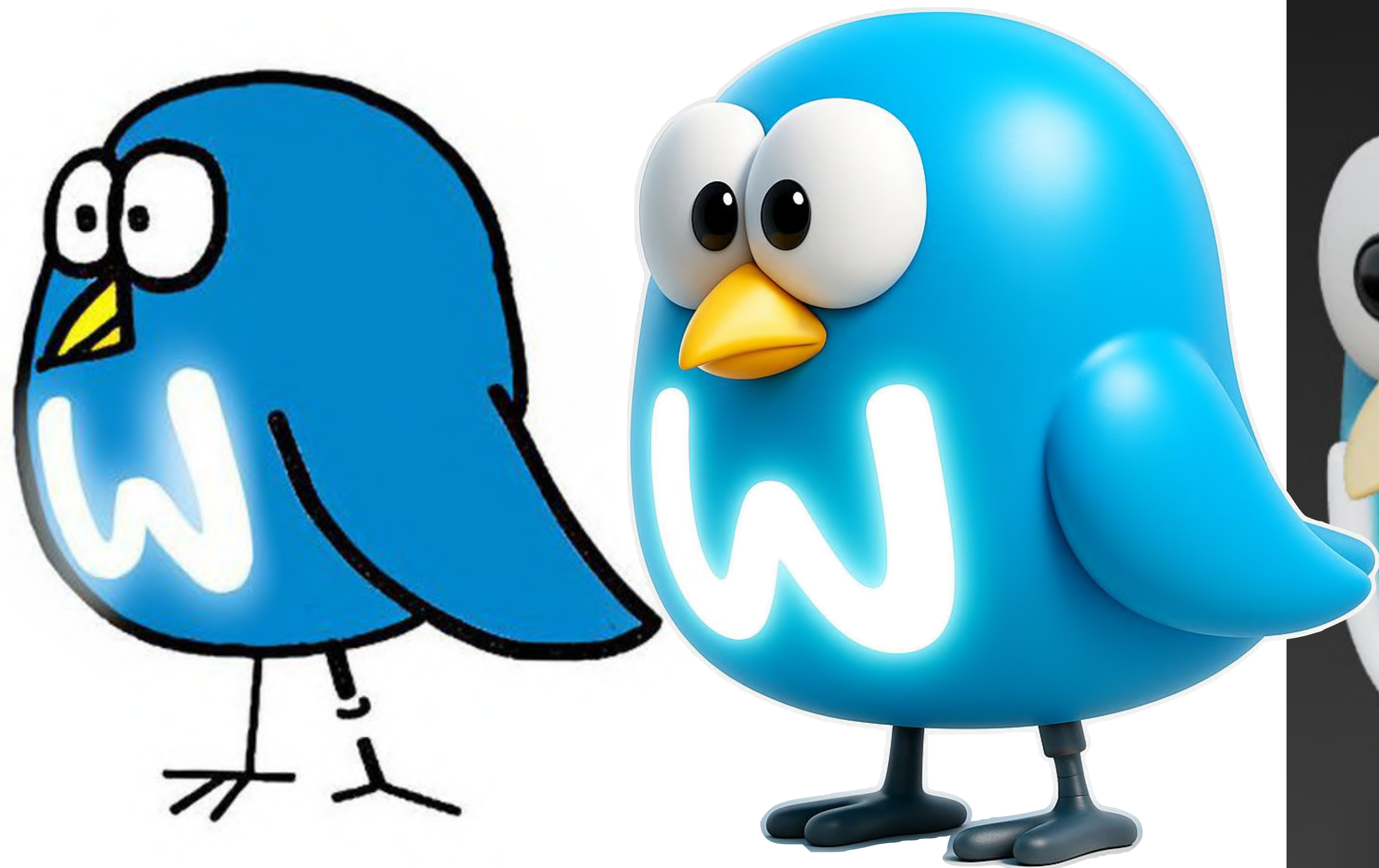


Latents Decoding



Real Use-Case

A Gift for a Friend

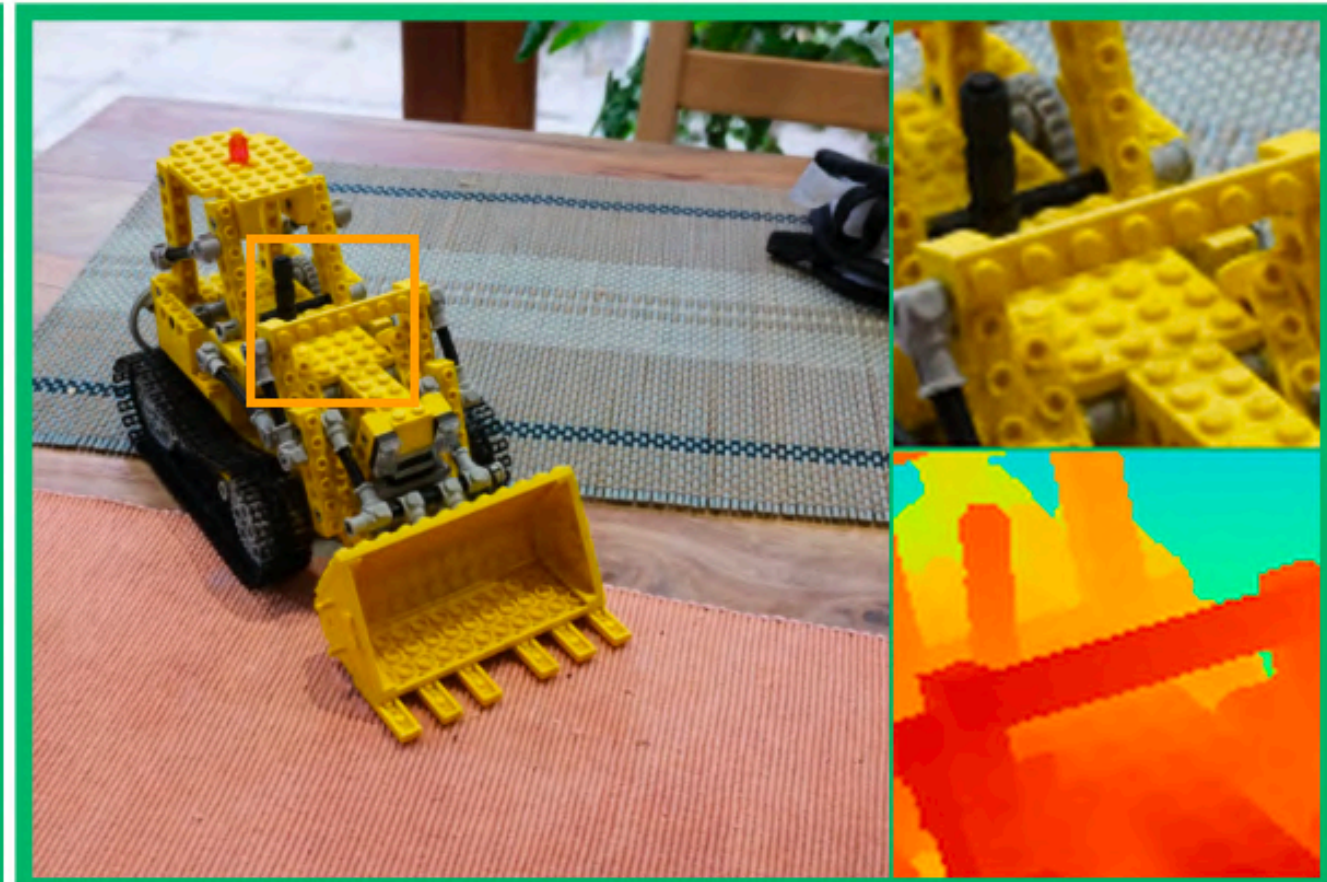
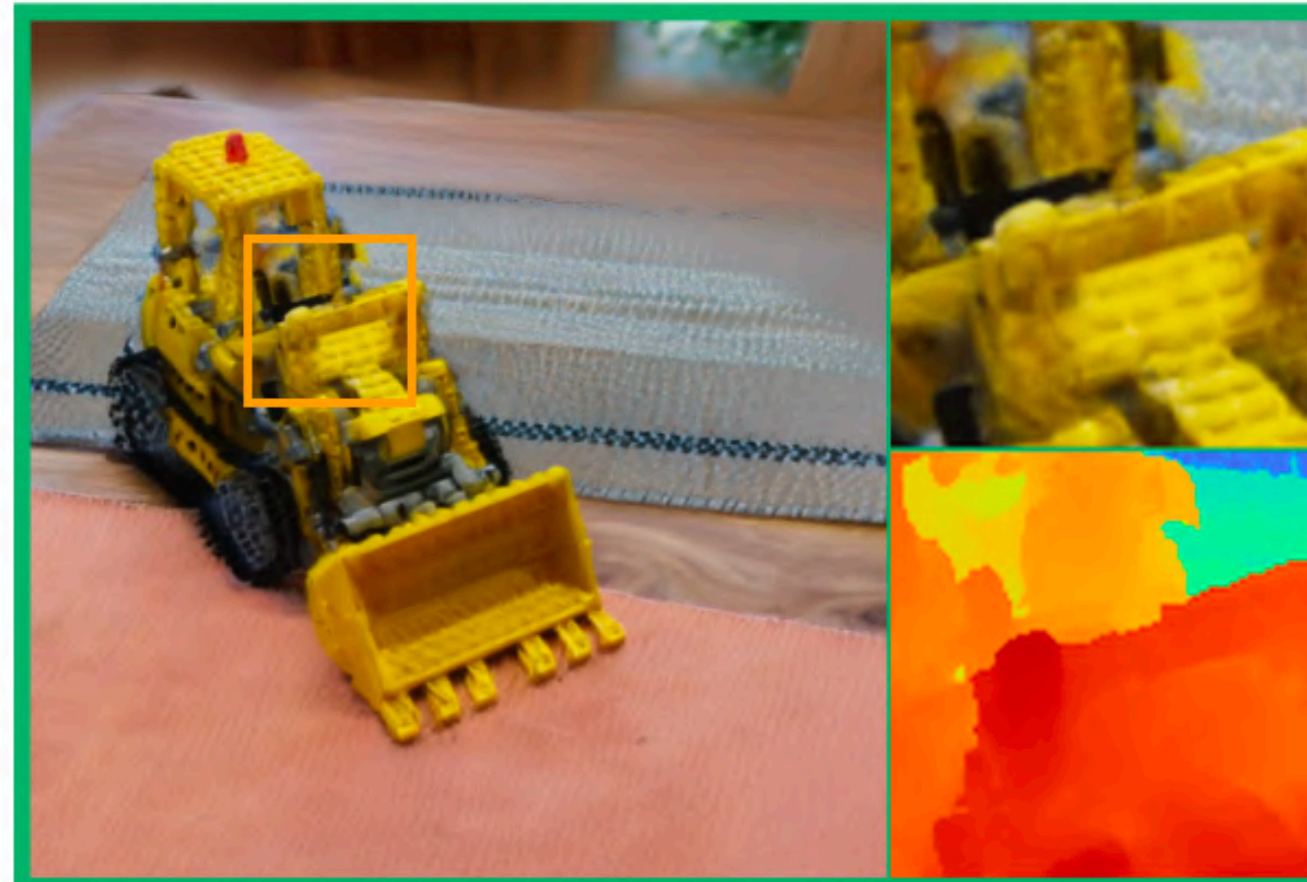


Fine-Tuning on Real Data

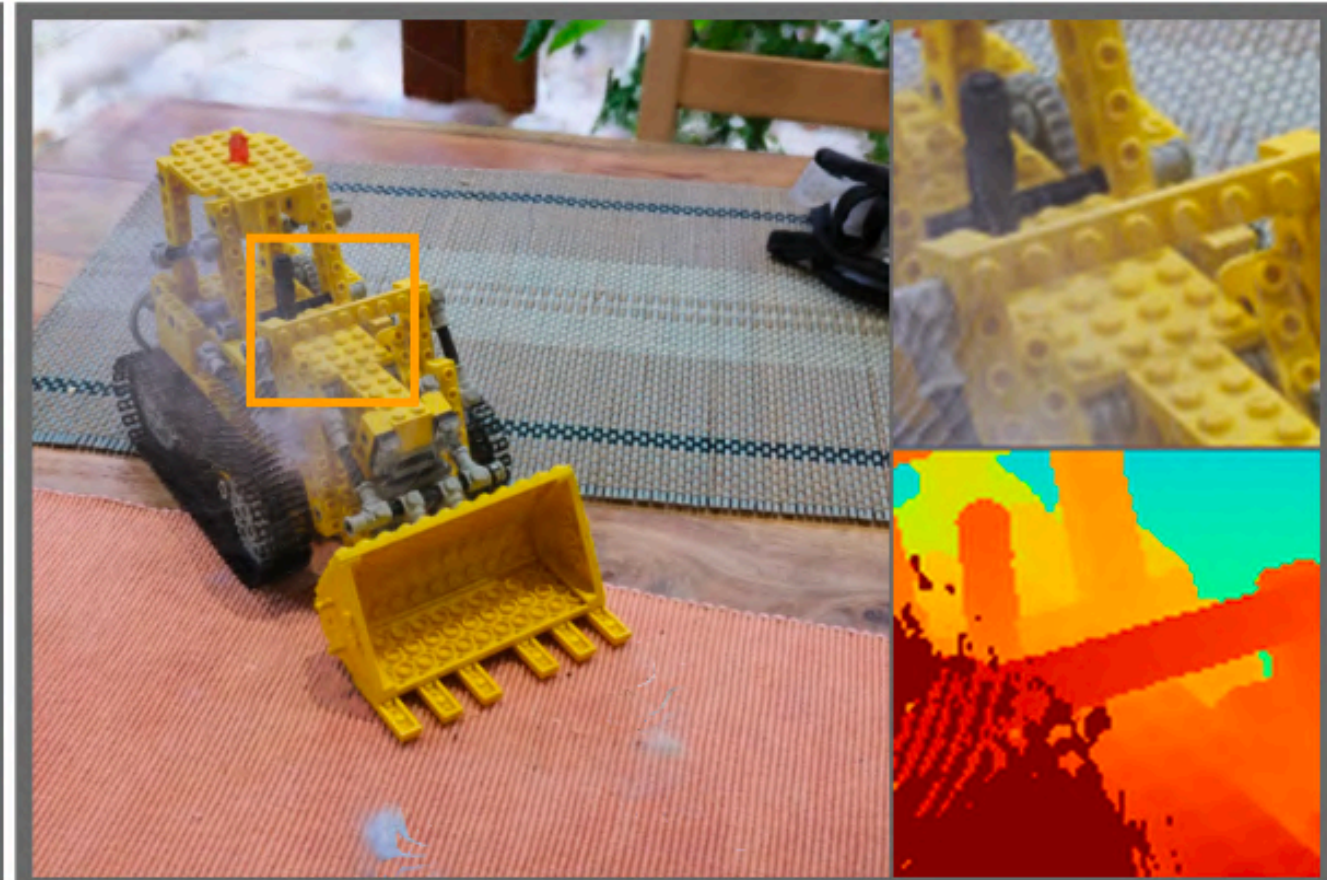
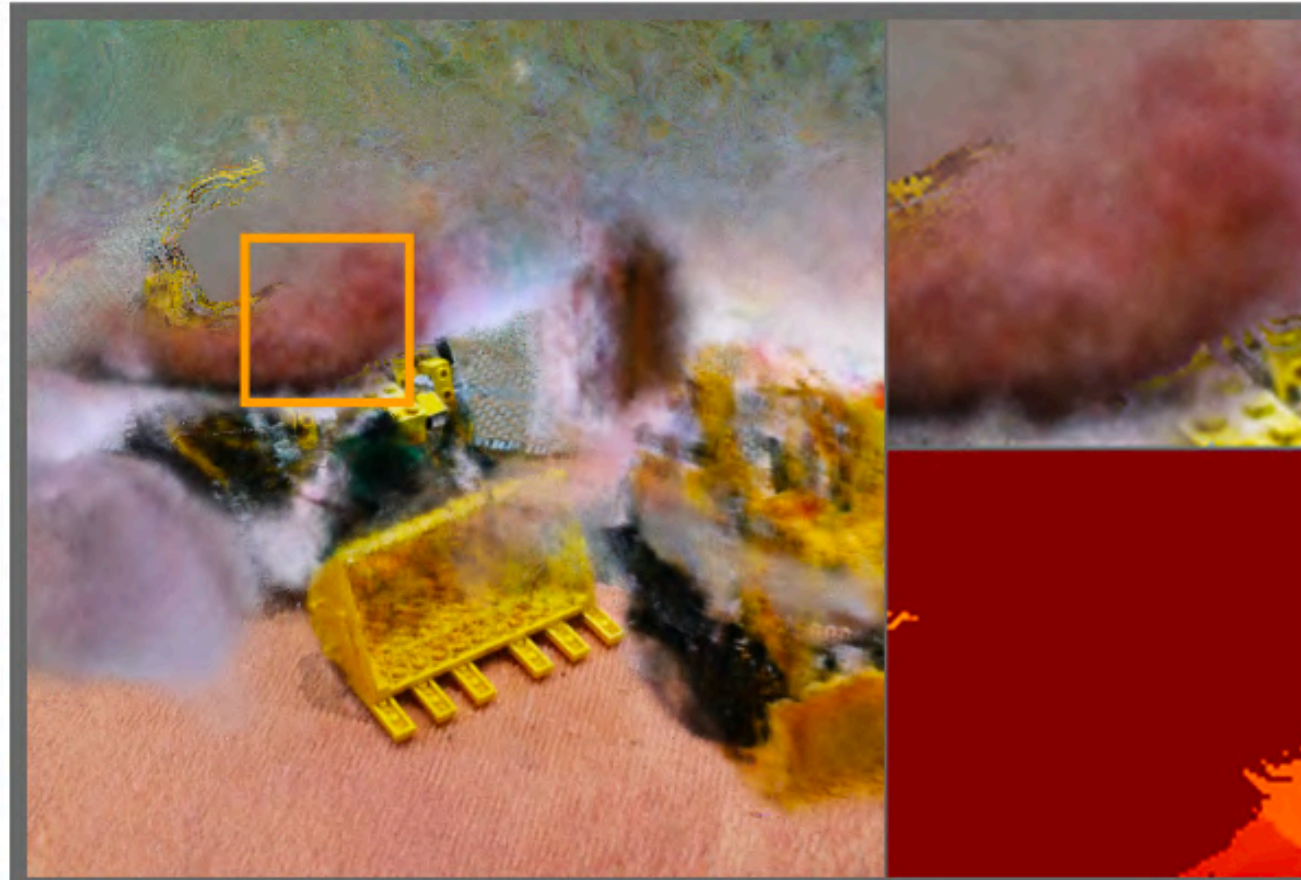
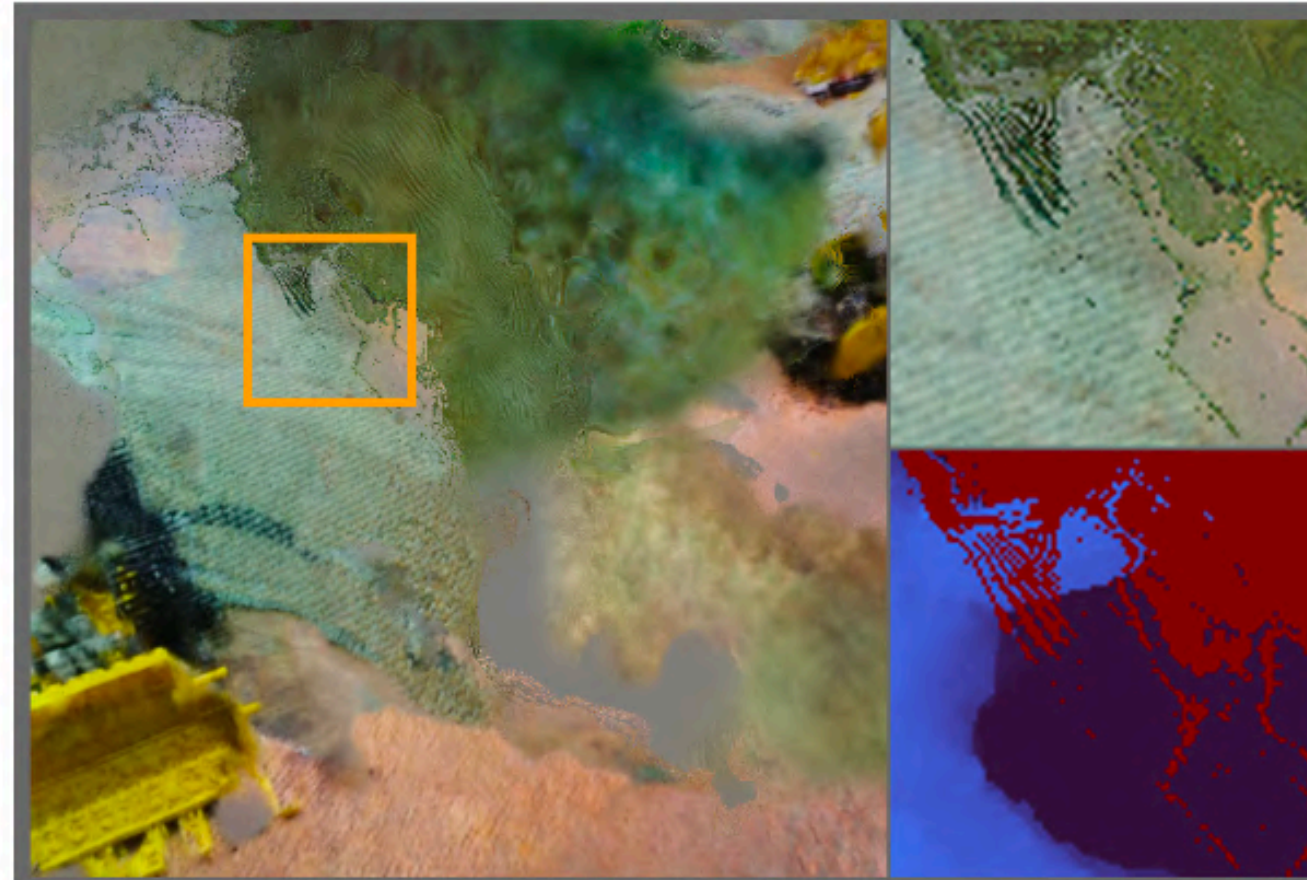


The Shift to Real Data

ReconFusion



Zip-NeRF [2]



3 views

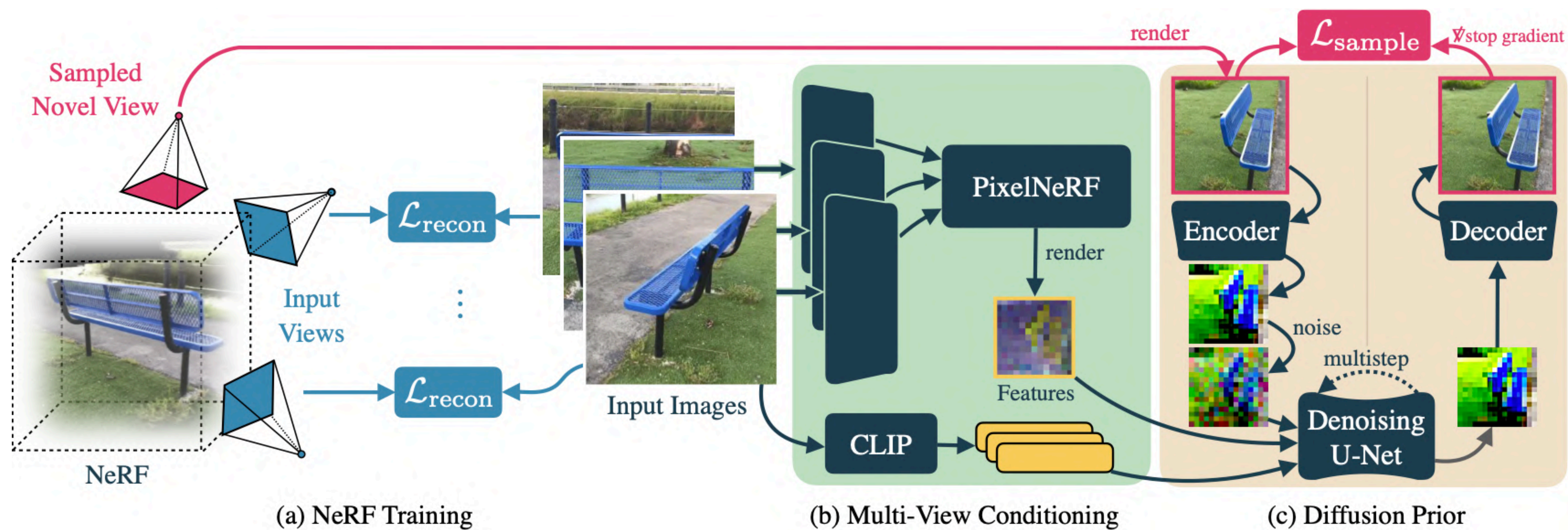
9 views

54 views

Training

- Initialise with a pretrained text-to-image model
- Fine-tune *collect as much data as you can*
 - Objaverse
 - CO3D
 - RealEstate10k
 - MVImageNet

ReconFusion



Conditional Generation Examples



(a) 3 input views



(b) 6 input views



(c) 9 input views

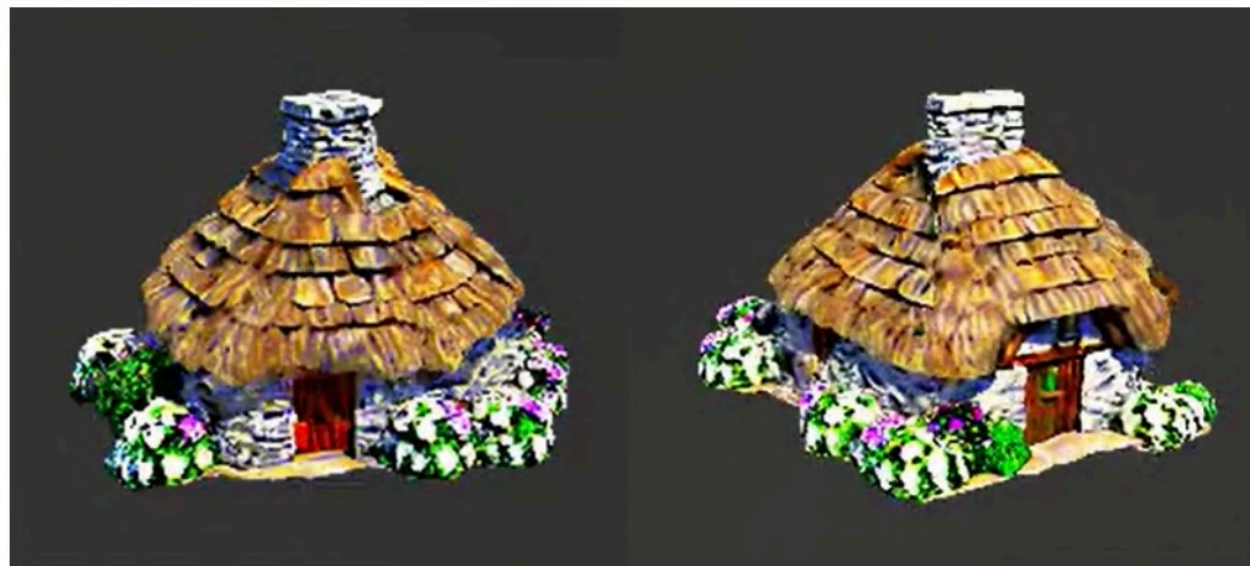
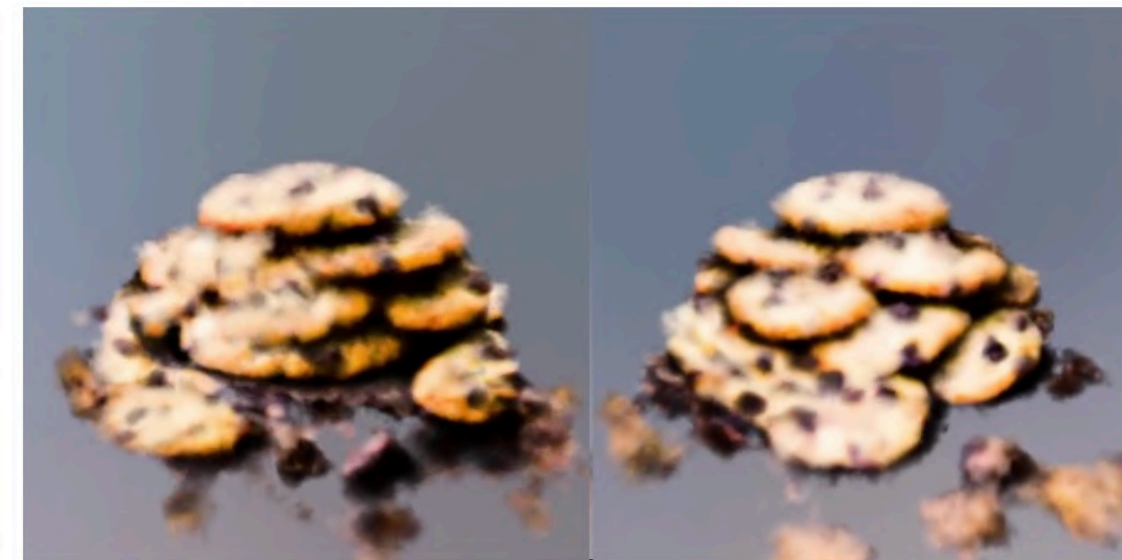
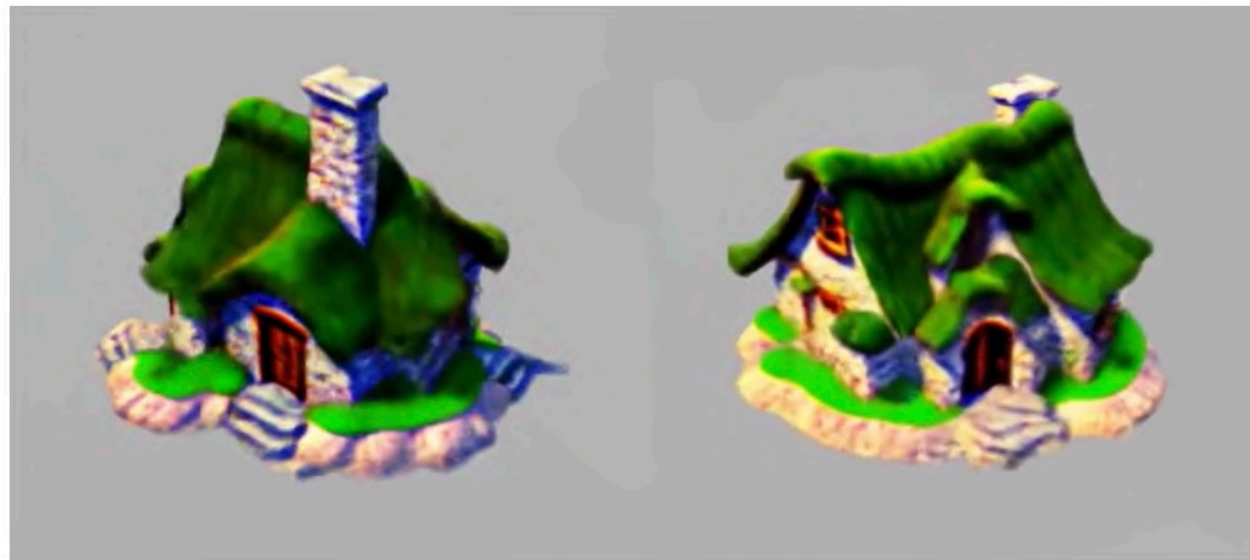


(d) Ground truth

Cross-View Consistency is Still an Issue

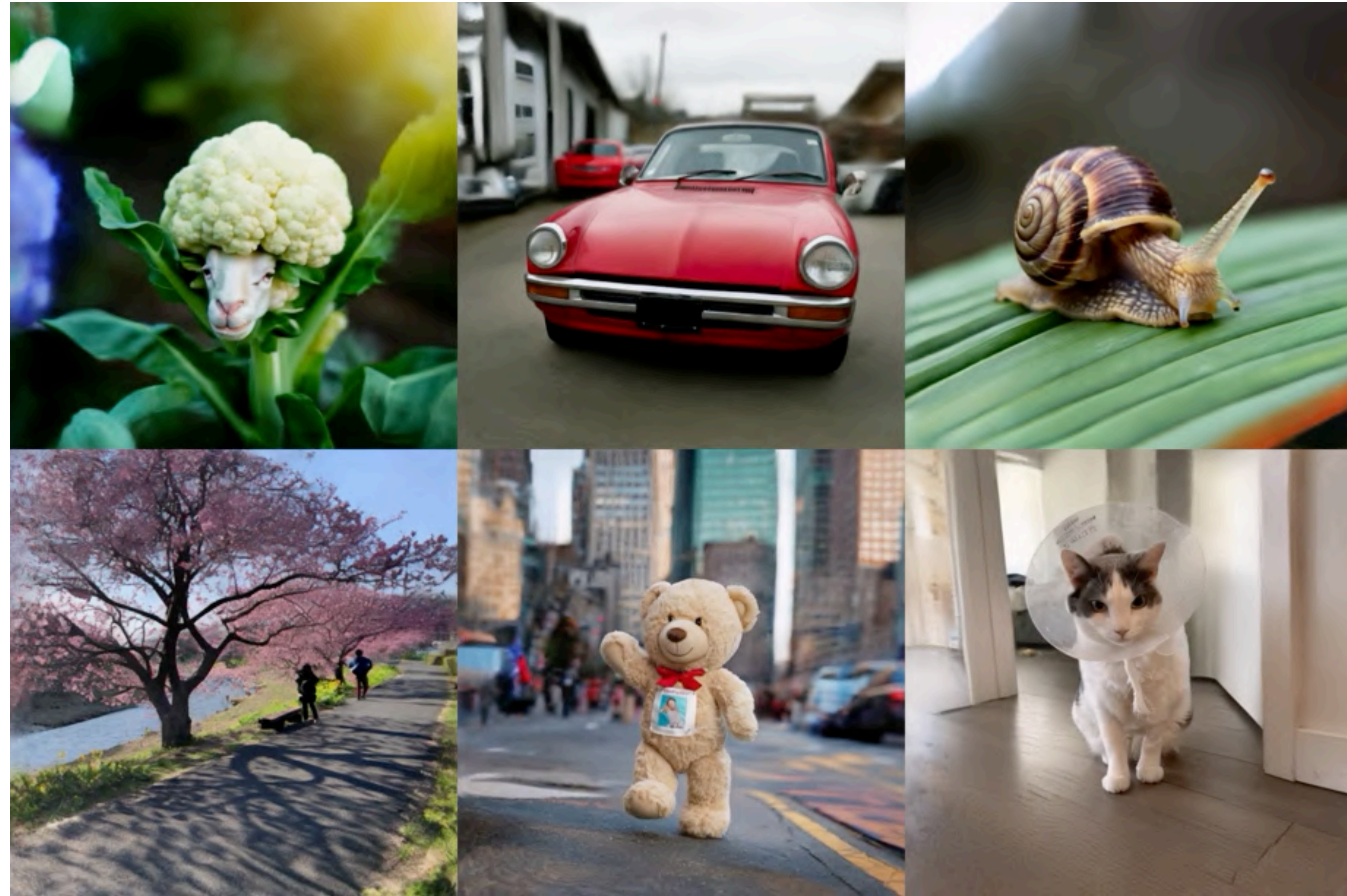


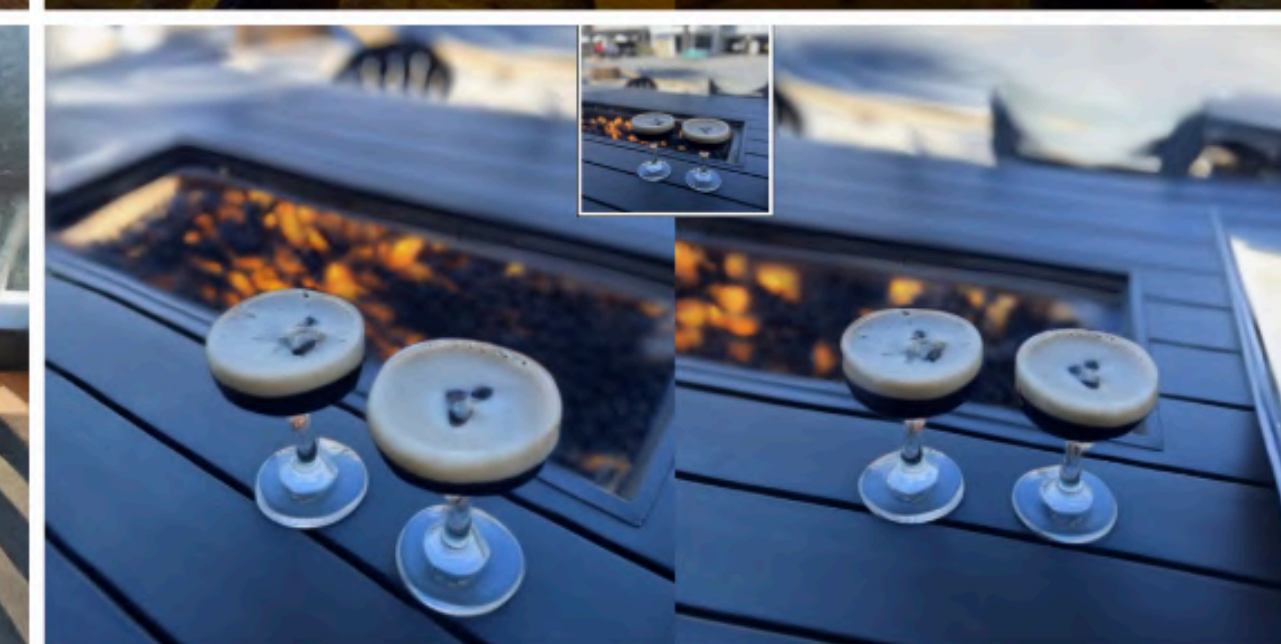
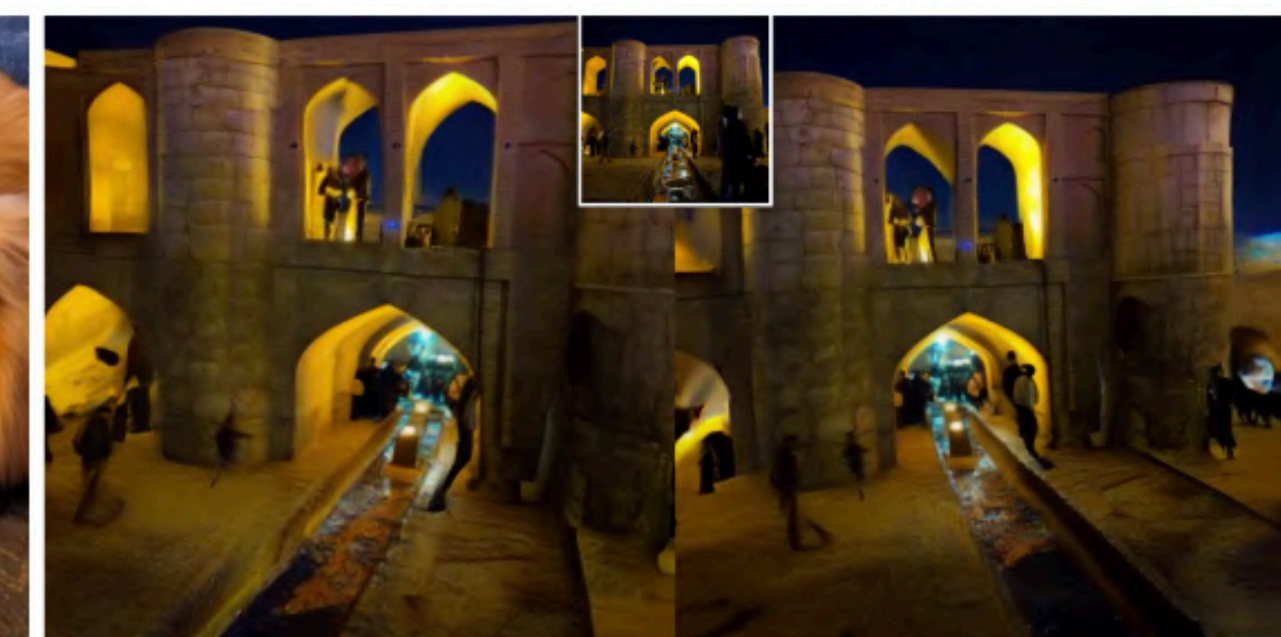
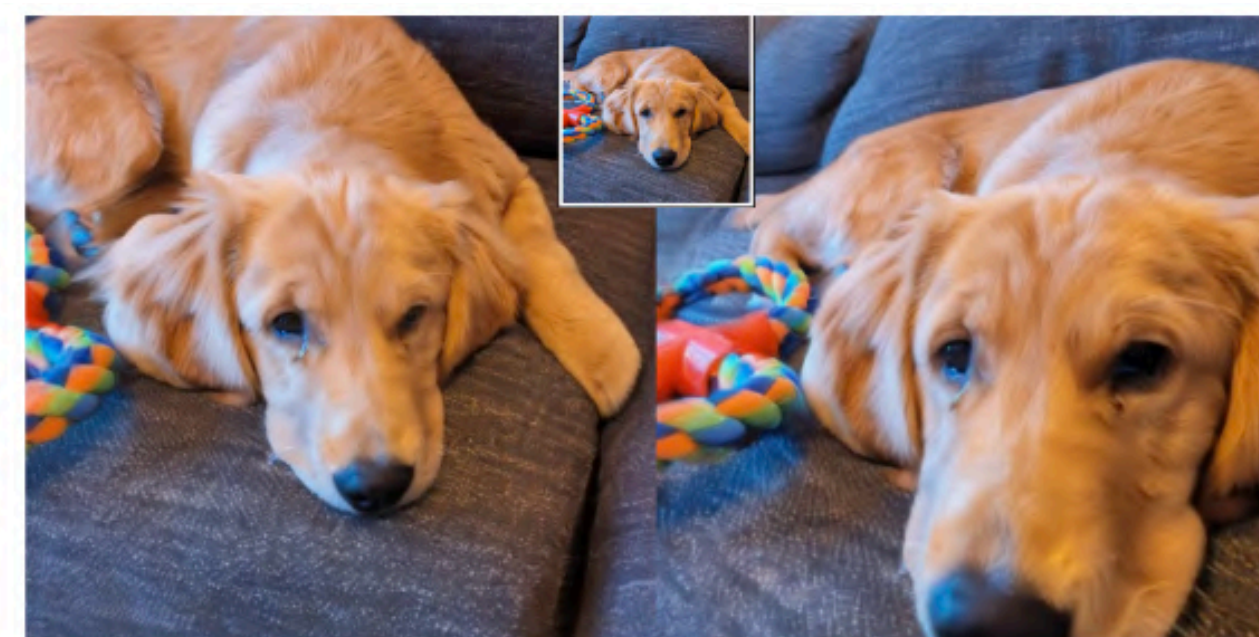
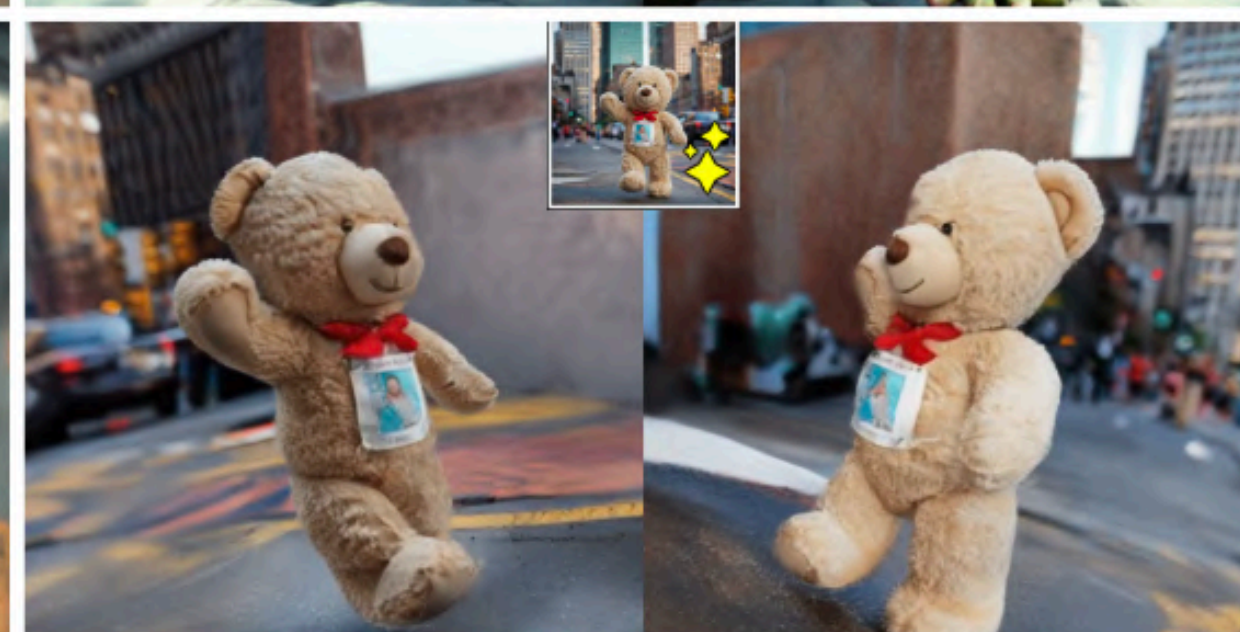
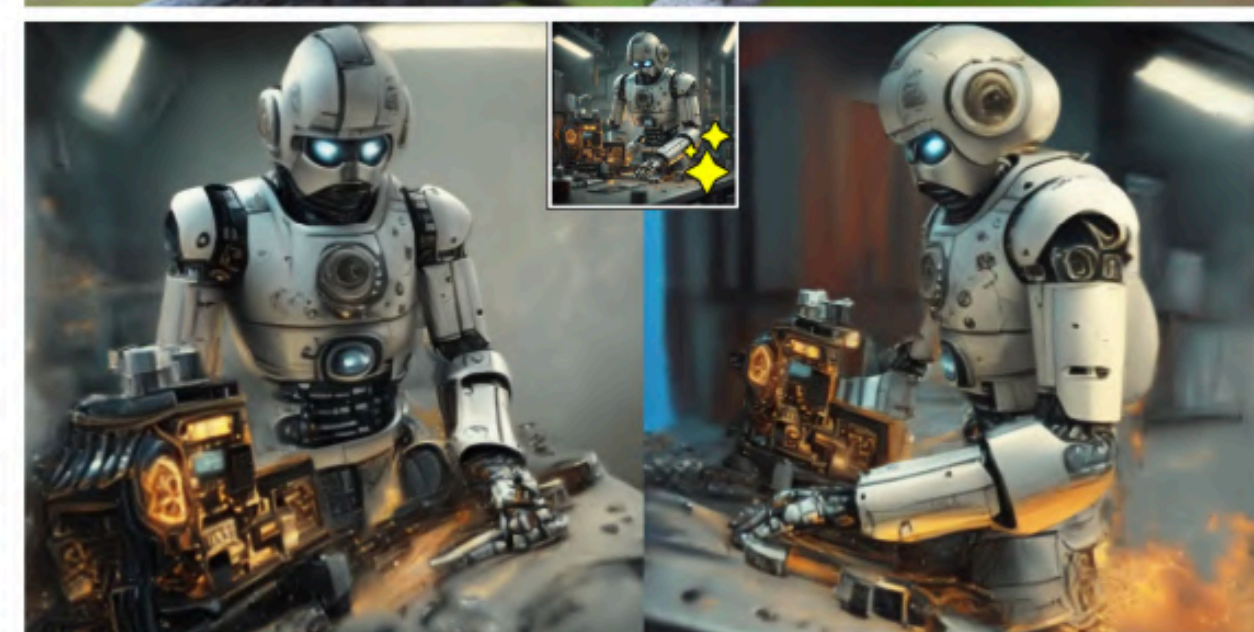
Sora Generates Videos with Stunning Geometrical Consistency



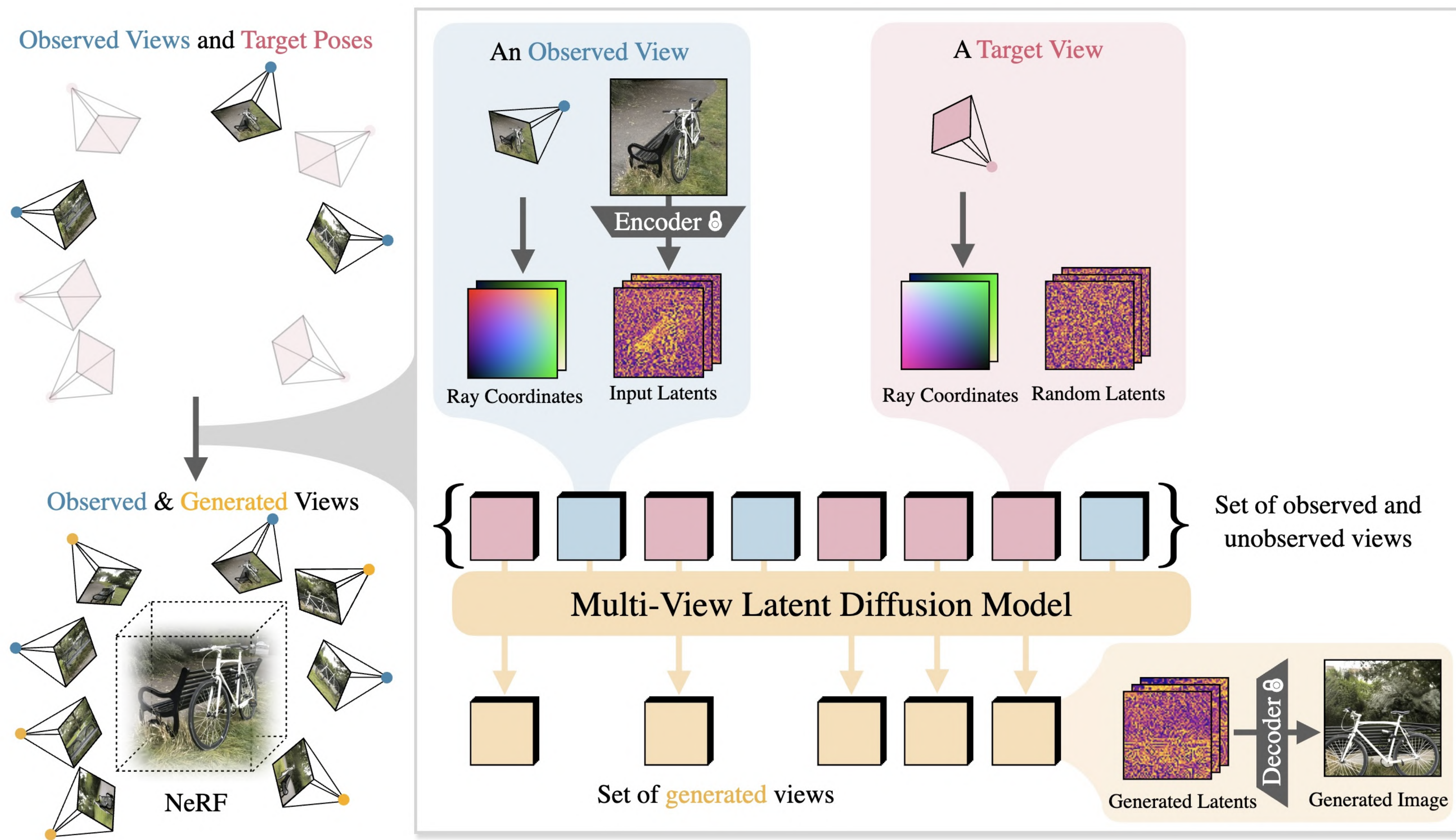
Cat3D

- Novelty
 - Simpler generation pipeline
 - Simpler diffusion conditioning
- Not novel, yet important:
 - Simultaneous generation of views
- Tasks
 - Text-to-3d, Image-to-3D, Novel View Synthesis

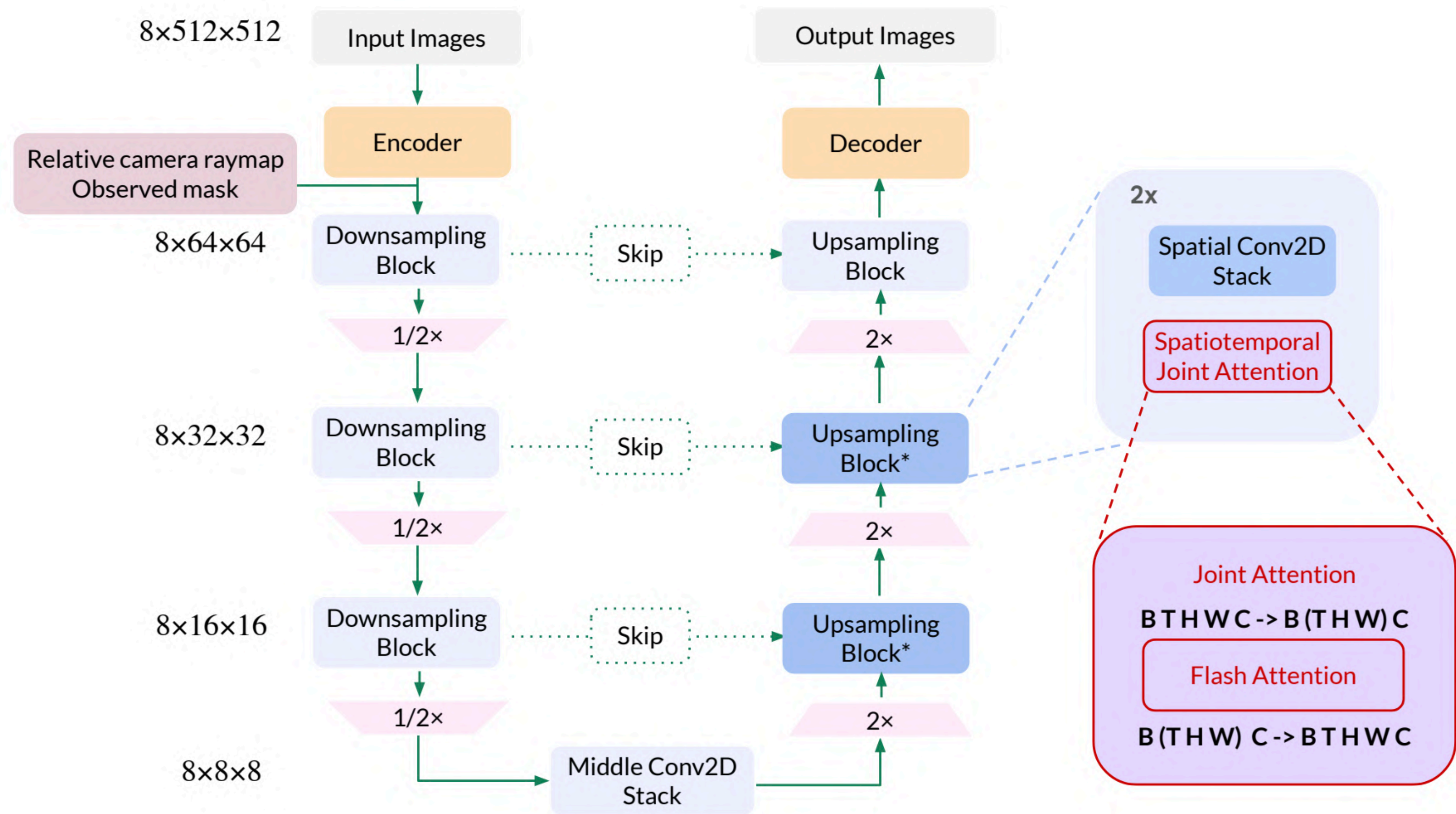




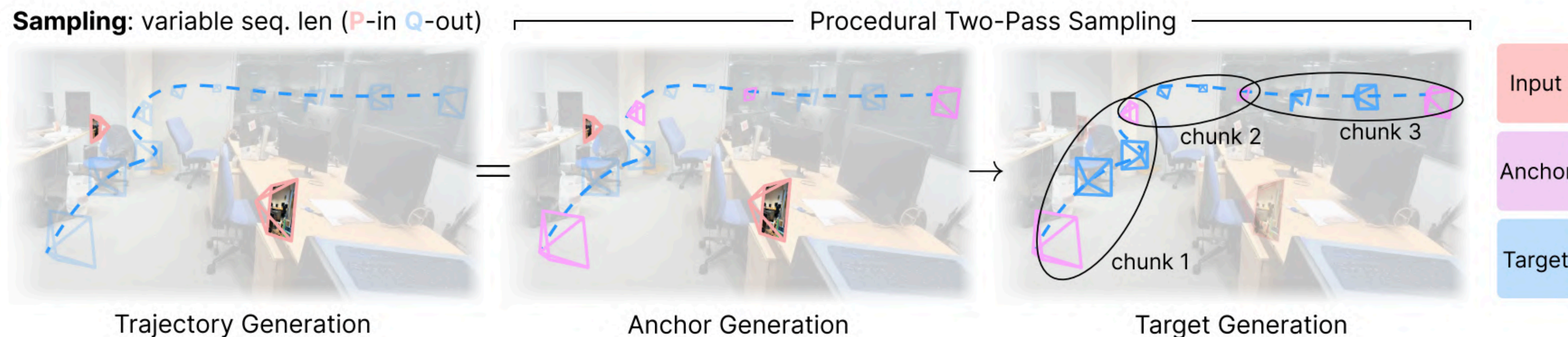
Overall Scheme for Data Generation



Diffusion Model Architecture



Temporal Consistency, No 3D Distillation, Open Source



Stable Virtual Camera

Single frame

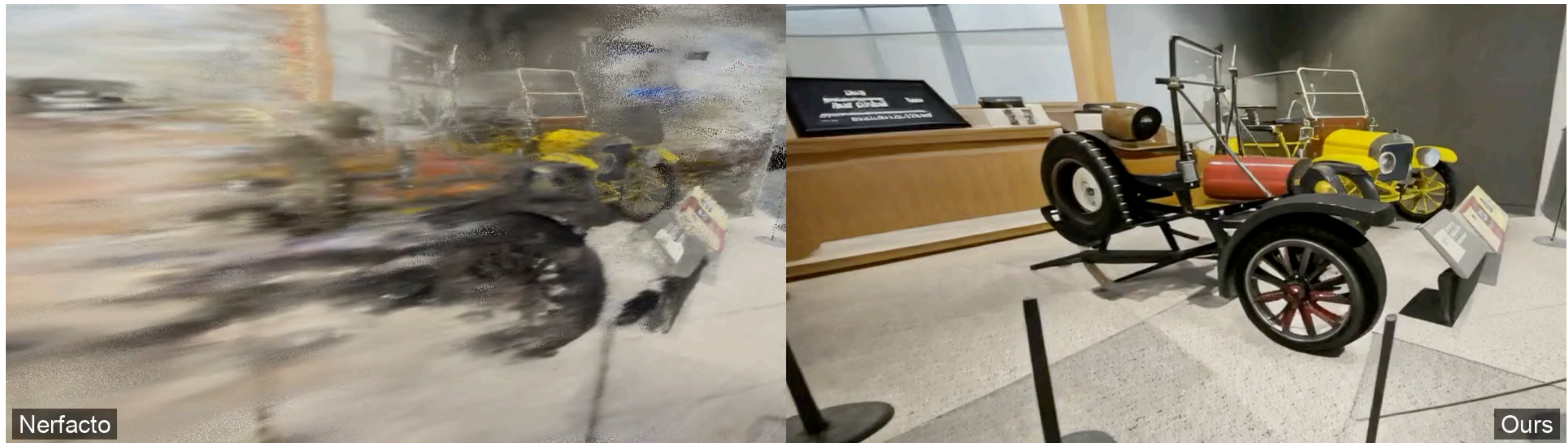
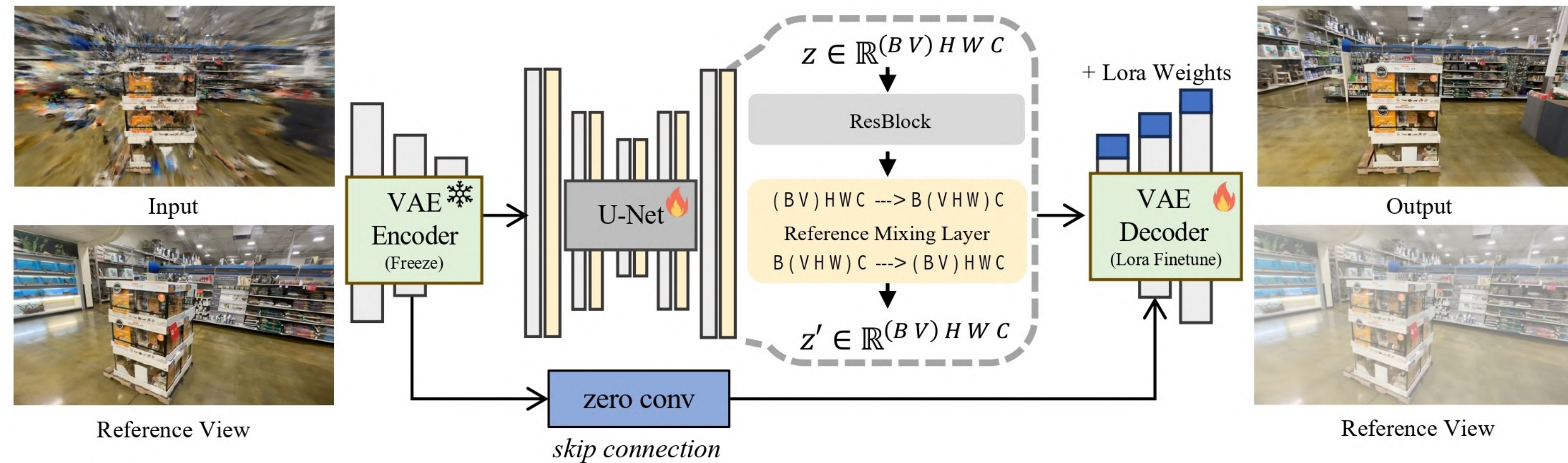


Multiple Frames:



Difix3D

Fix Splats Instead of Generating Data from Scratch



Video Models

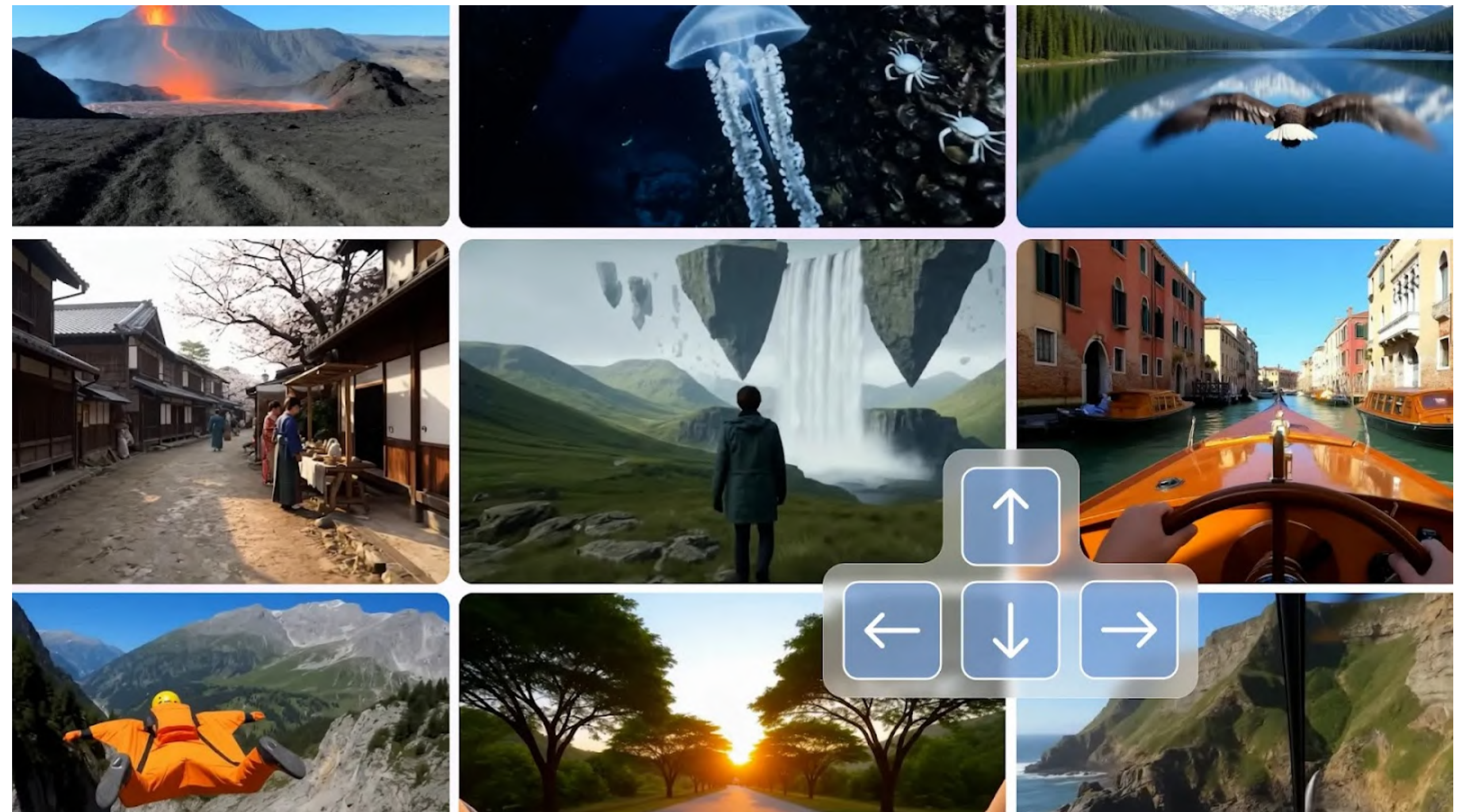
The Fundamental Limitation of NVS

- Novel View Synthesis assumes a static scene captured from multiple viewpoints
 - Expensive to collect, fragile
- Meanwhile, video data is abundant and captures the world in all its complexity
- In AI, scaling simple models on abundant data sooner or later beats methods relying on prior knowledge in the data (i.e., static world, calibrated camera)

World Models

A.k.a. Glorified Conditional Video Diffusion

- In short scenes beats model NVS-based methods in realism
- Lacks long-scale temporal consistency (at least for now)

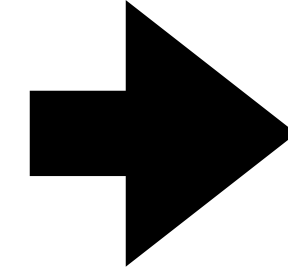


The Role of 3D in the Age of World Models

- If video models can already generate worlds, why do we still need explicit 3D?
- **Splats as an interface:** instead of generating a separate stream per user, generate a 3D representation once
- 3D knowledge helps with memory and consistency: **retrieval-augmented generation** with 3D data



Summary



What if it were **twilight** on this street?

- Data in 3D is scarce
- Diffusion models can be leveraged for 3D generation
 - Score Distillation Sampling does this out-of-the-box
 - Training from scratch on synthetic 3D data
 - Fine-tuning on real data helps with NVS
- World models take center stage, 3D is crucial for consistency & interactions